

Problem Set: Causality and Potential Outcomes

BUSS975: Causal Inference in Financial Research

Four problems. Plan on about three hours. Problems 3(c) and 4(a)–(c) ask you to compute in R (or Stata); attach your code and output. Nothing else requires software.

Where a part asks you to explain or evaluate, **answer in at most three sentences**. These are graded on precision, not length.

Further practice — including the regression-bridge material cut from this set — is in the companion file `01_causality_PS_extra.qmd`. It is optional, and good preparation for the comprehensive exam.

Problem 1 (Assignment mechanisms and the SDO decomposition)

What this is for: everything in this course rests on the claim that the assignment mechanism, not the estimator, determines whether a comparison is causal. This problem makes that claim arithmetic.

A banking regulator can inject capital into distressed banks. Let $D_i = 1$ if bank i receives an injection, and let the outcome Y_i be lending growth (%) over the following year. The complete (and, in reality, unobservable) potential outcomes for ten banks are:

Bank i	1	2	3	4	5	6	7	8	9	10
Y_i^0	1	2	3	2	4	7	8	9	8	10
Y_i^1	8	8	8	6	7	9	9	9	7	8

Regulator A has perfect foresight and injects capital into the five banks with the **largest treatment effects**.

- Compute δ_i for each bank and identify the five treated under Regulator A. Compute the true ATE , ATT , and ATU , and verify that $ATE = \pi \cdot ATT + (1 - \pi) \cdot ATU$, where π is the share treated.
- Write out the observed data table — the Y_i an econometrician actually sees, together with D_i — and say which entries of the potential outcomes table are now missing. Compute the simple difference in outcomes,

$$SDO = E[Y \mid D = 1] - E[Y \mid D = 0].$$

- (c) Compute the selection bias term and the heterogeneous treatment effect bias term, and verify numerically that

$$SDO = ATE + \underbrace{\left(E[Y^0|D=1] - E[Y^0|D=0]\right)}_{\text{selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{heterogeneous TE bias}}.$$

- (d) **Regulator B** instead injects capital into banks 6–10, the five healthiest — perhaps because they are the most politically connected. Recompute ATT , ATU , and SDO . Present the true ATE and the two $SDOs$ in a small table. Then explain why ATE was unchanged across the two regulators while ATT was not.
- (e) Under Regulator A the true ATE is positive but the SDO is negative. Explain — as you would to a journalist — why a well-intentioned, *competent* regulator produces data suggesting its own program was harmful. Then evaluate a colleague’s proposal: “the problem is that the sample is too small; with 10,000 banks the bias would disappear.”

Problem 2 (SUTVA in financial markets)

What this is for: SUTVA is the assumption that is violated most often in finance and discussed least often. It is also the one that no amount of data or cleverness with estimators can repair.

- (a) For each scenario, state whether the violation is of **no interference** or **no hidden variation in treatment**, explain the mechanism in one sentence, and state whether the naive treated-minus-control comparison would likely **overstate** or **understate** the true effect of treatment on the treated.
- A government recapitalizes half the banks in a concentrated banking market. Treated banks use the capital to bid aggressively for deposits and creditworthy borrowers, drawing both away from untreated banks.
 - A researcher studies “the effect of receiving TARP funds” on bank lending. TARP investments in the sample range from \$200 million to \$25 billion, and larger banks received larger injections.
 - A subsidy is given to a random 30% of firms in an industry. The industry’s total output is fixed by demand, so subsidized firms gain market share exactly at the expense of unsubsidized firms.
 - A study estimates the effect of “adopting an anti-takeover provision” on firm value, pooling classified boards, poison pills, and supermajority requirements into a single indicator.
- (b) In scenario (iii), the researcher’s difference-in-differences finds a large positive effect on treated firms’ output. Is the estimate *wrong*? Distinguish carefully between the **partial-equilibrium** and **general-equilibrium** effects, and state which question each answers.
- (c) Scenario (ii) can be repaired by redefining the treatment. Propose a redefinition, and state precisely what estimand you are left with and how it differs from the one the researcher wanted.
- (d) Explain why SUTVA is logically **prior** to the independence assumption — that is, why a SUTVA violation is a more fundamental problem than selection bias.

Problem 3 (Randomization inference: the exact case)

What this is for: finance is full of settings with one treated state, one regulatory event, or five treated firms. Asymptotic standard errors are meaningless there. This is the alternative, and it is exact.

Six firms are enrolled in a pilot program. Three are selected **by lot** to receive treatment. The realized assignment and observed outcomes are:

Firm i	1	2	3	4	5	6
D_i	1	1	1	0	0	0
Y_i	6	7	8	3	4	5

- (a) Compute the observed SDO . State Fisher's sharp null hypothesis, explain how it differs from the null that the *average* treatment effect is zero, and say why the sharp null — but not the average null — allows you to fill in every missing potential outcome.
- (b) How many distinct assignments of three treated firms out of six are possible? Under the sharp null, explain why the observed Y_i would have been identical under every one of them. Without enumerating, identify the assignment producing the largest SDO and the one producing the smallest, and say where the realized assignment sits in that distribution.
- (c) **Computational.** Compute the exact two-sided p -value, defined as the share of assignments for which $|SDO| \geq |SDO_{\text{observed}}|$. A skeleton is given below; report your code and output.

```
y <- c(6, 7, 8, 3, 4, 5)
treated <- c(1, 2, 3)
sdo_obs <- mean(y[treated]) - mean(y[-treated])

perms <- combn(6, 3)           # all possible assignments (columns)
sdo_all <- apply(perms, 2, function(tr) mean(y[tr]) - mean(y[-tr]))

mean(abs(sdo_all) >= abs(sdo_obs))  # exact two-sided p-value
```

- (d) A colleague suggests: “Rather than all this permuting, just bootstrap the SDO — resample the six firms with replacement a few thousand times and read off the interval.” Explain what the bootstrap resamples and what randomization inference permutes, state which source of uncertainty each one describes, and say which is the right object here given how these firms were selected into treatment.
- (e) The code uses `combn(6, 3)`, which presumes the experiment fixed **exactly three** treated firms. Suppose instead the researcher had assigned each firm independently by a fair coin flip. How many assignments are in the reference set now? Explain why `combn(6, 3)` would then be the wrong denominator, and state in one sentence what this says about the relationship between the permutation scheme and the physical randomization.
- (f) The realized assignment was the most extreme of all possible assignments, yet the p -value does not fall below 0.05. Explain what this implies about the power of very small experiments, state the smallest attainable two-sided p -value with six units and three treated, and name one method later in the course that relies on exactly this logic.

Problem 4 (Choosing a test statistic, and when exact enumeration fails)

What this is for: the sharp null tells you what to hold fixed, but it does not tell you what to measure. The test statistic is a separate choice, and it decides what you are able to detect.

- (a) Return to the six firms of Problem 3. Compute the **normalised rank** statistic, where $\tilde{R}_i = \sum_j I(Y_j \leq Y_i) - (N + 1)/2$ and the statistic is the difference in mean $\tilde{R}$ between treated and control. Compute also the **Kolmogorov–Smirnov** statistic, $T_{KS} = \max_y |\hat{F}_T(y) - \hat{F}_C(y)|$. Obtain the exact two-sided p -value for each. You should find that SDO , the rank statistic, and T_{KS} all deliver the **same** p -value. Explain why — what feature of this particular dataset forces the three to agree, and what is actually binding on the p -value here?
- (b) A fourth candidate is the log difference,

$$T_{\log} = \frac{1}{N_T} \sum_i D_i \ln Y_i - \frac{1}{N_C} \sum_i (1 - D_i) \ln Y_i.$$

Suppose the treatment scales every firm’s outcome by a common factor, $Y_i^1 = c Y_i^0$. Show that the per-unit effect in **levels** depends on Y_i^0 but the per-unit effect in **logs** does not, and explain in one sentence why this makes $T_{\log}$ the better-powered statistic against multiplicative alternatives. Then state why $T_{\log}$ cannot be used at all on the data in part (c).

- (c) **Computational.** A regulator randomises a disclosure requirement across 60 listed firms, 30 treated. The outcome is the firm’s abnormal return (%) over the following quarter. Generate the data exactly as below.

```
set.seed(3)
control <- round(rnorm(30, 0, 1), 2)
treated <- round(rnorm(30, 0, 4), 2)
y <- c(control, treated)
d <- c(rep(0, 30), rep(1, 30))
```

- Verify that $\binom{60}{30} \approx 1.2 \times 10^{17}$, and explain in one sentence why this rules out the approach you used in Problem 3(c) and what you must do instead.
 - Compute the **approximate** two-sided p -value for the SDO using 10,000 random reassignments that keep 30 firms treated. Report it.
 - Compute the approximate p -value for T_{KS} on the same 10,000 draws. Report it.
 - The two p -values are very different. Report the treated and control **standard deviations**, and explain what the disclosure requirement did to the distribution of abnormal returns that a difference in means was structurally unable to see. State the economic interpretation: what has the regulator changed, if not the average?
- (d) Having found nothing with the SDO in (c.ii), a researcher tries T_{KS} , finds (c.iii), and reports only the second. Explain precisely what is wrong with this, and state the practice that would have made the T_{KS} result credible. In doing so, be careful to distinguish the **validity** of a randomization test from its **power** — say which of the two the choice of test statistic affects, and which it does not.